

Graphing $f(x) = |x|$ and Key Attributes - Practice

Algebra II | Unit 2 | Lesson 1 Practice | TEK A2.2.A

Objective: I can evaluate, graph, and state the key attributes of $f(x) = |x|$, and compare it to a linear function.**Section 1 — Recall**

1. Evaluate $f(-9)$, $f(5)$, and $f(0)$ for $f(x) = |x|$.

 $f(-9) = \underline{\hspace{2cm}}$ $f(5) = \underline{\hspace{2cm}}$ $f(0) = \underline{\hspace{2cm}}$

2. State the domain, range, and vertex of $f(x) = |x|$.

Domain: _____ Range: _____ Vertex: _____

Section 2 — Apply

3. Complete the table for $f(x) = |x|$, then state whether the graph is symmetric, and about what.

x	-4	-1	0	2	5
f(x)					

Symmetric? _____

4. The line $y = x$ and $f(x) = |x|$ share the same domain. Explain why their ranges differ.

Section 3 — Challenge

5. Points $(-8, 8)$ and $(8, 8)$ both lie on $f(x) = |x|$. Give a DIFFERENT pair of points with matching y-coordinates, and explain the pattern.

New pair (sample): _____

6. A student says $f(x) = |x|$ has two x-intercepts because it is V-shaped, like some parabolas they have seen. Is this true? Justify using the vertex.

Section 4 — Explain It

7. Explain in your own words why the range of $f(x) = |x|$ is $[0, \infty)$ and can never include a negative number. Connect your answer to the real-world "distance" meaning from today's lesson.
